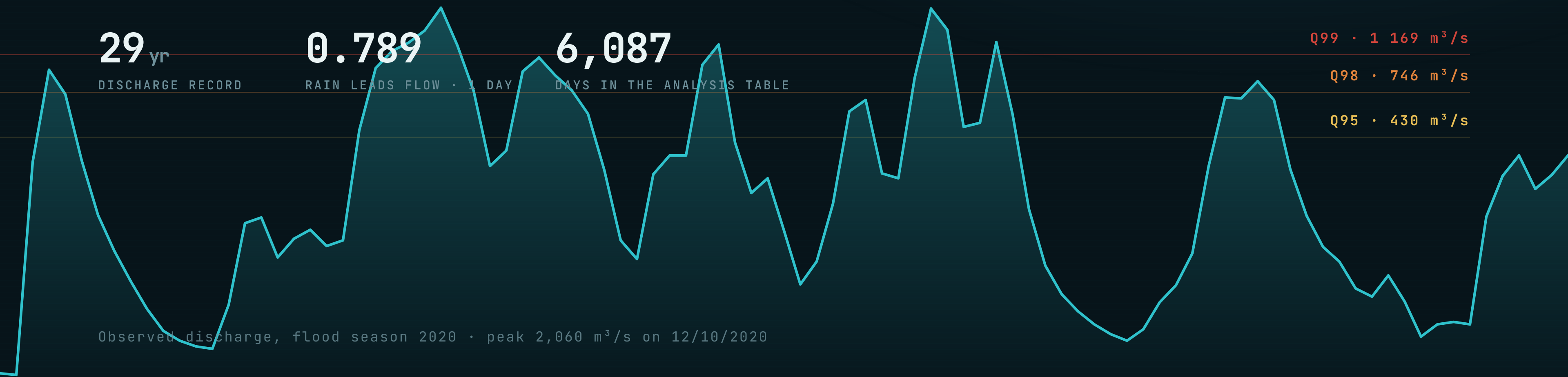


Measured, not trusted

How the open data were collected, cleaned and explored – and what we corrected since Report 1.



Four open sources — every one measured, not trusted

River discharge

GloFAS v4, ~5 km grid, daily. Documented from 1984 — **values only from 1997**.

Rainfall

ERA5 reanalysis, **64 grid cells**, daily 2010–2026, hourly 2015–2026.

Sea level

Marine API at the river mouth — a tide proxy, **only from 2023**.

Ground truth

19 documented floods + official alert stages from Decision 05/2020/QD-TTg.

All free, no API key, licence **CC BY 4.0**. Raw store: **1,295 files · 98 MB** → one table of **6,087 days × 71 columns**.

Probe cheaply, then fetch only what matters

361

cells probed with one year of data

234

returned any flow

83

full 1997–2026 series

1

target

Rejected: the **largest-flow cell** (it merges the Huong and Bo rivers) and the **cell nearest the gauge** (the model's channel is ~7 km away). Chosen by **catchment area implied by runoff**.

Four defects — each passed the obvious check

13 empty years

1984–1996 delivered as nulls in all 83 cells. Acceptance counted **rows, not values**.

Uneven rain grid

Middle sub-basin — nearest the forecast point — had **9 of 16 cells**. Rebuilt to 64 uniform.

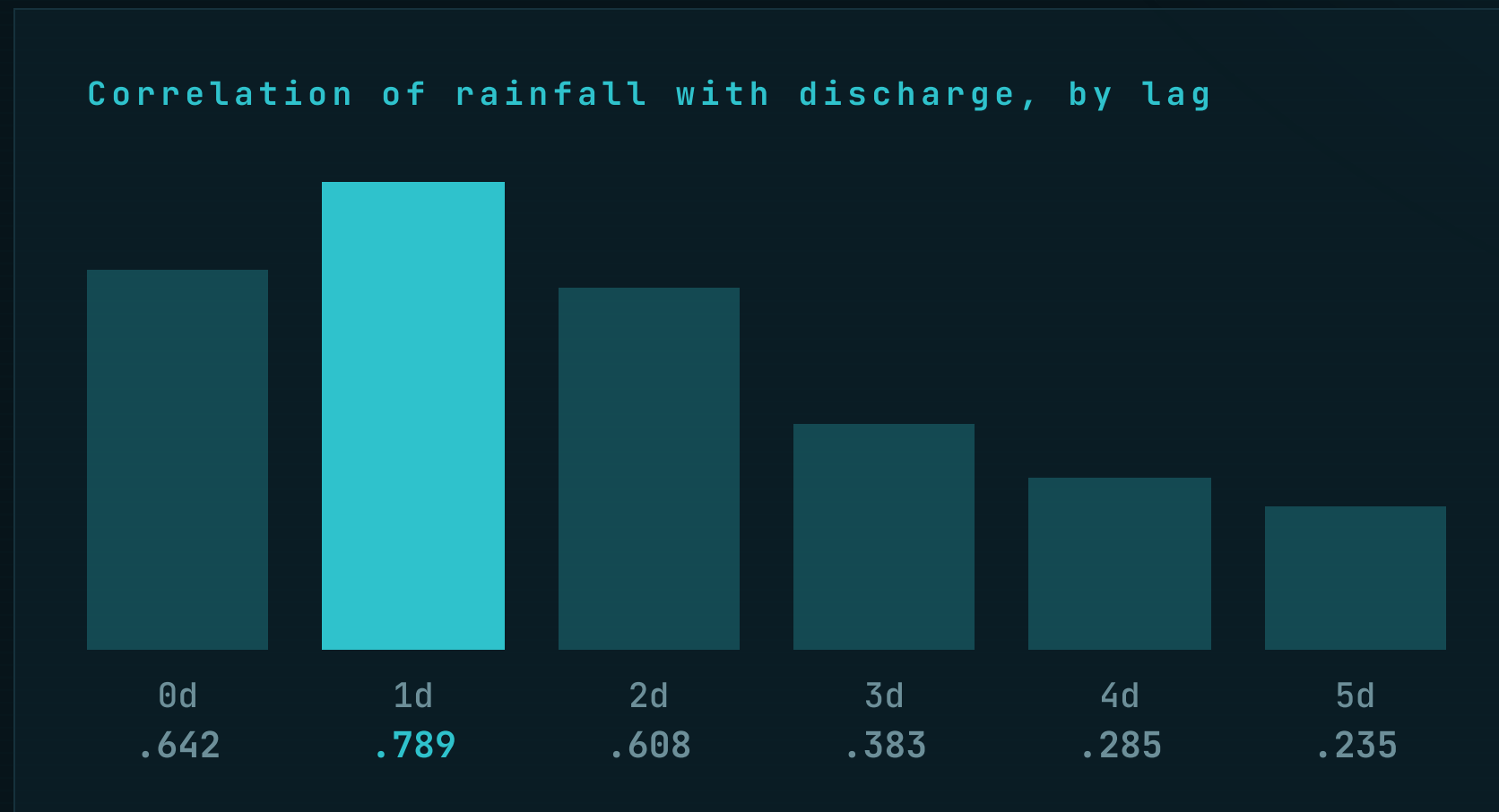
“Forecast” = observed

Archived forecast rain identical to ERA5: **max difference 0.0 mm** over 1,523 days.

Wrong coordinate

Planned point gave **5.7 m³/s**; the river carries ~308.

Rain today, river tomorrow



- Peak at **one day**, same in all three sub-basins — travel time is only **5–6 hours**.
- **3-day accumulated rain** is the strongest rain feature: **$r = 0.863$** .
- Same heavy rain, wet vs dry ground: **585 vs 85 m^3/s** — a factor of **6.9**.

Water level does not follow flow

4.17 m

Oct 2020
2,060 m³/s

4.34 m

Nov 2023 – higher level
only **584 m³/s**

61

days above the 99th
percentile in 16 years

At Kim Long, log-flow explains little of the published level ($R^2 = 0.18$). Tide and backwater from the lagoon are likely causes – so flood risk is defined by **discharge percentiles**, not official stages.

— LIMITATIONS — STATED NOW

What the data cannot tell us

- Discharge is **simulated, not measured** — no gauge record.
- ERA5 **under-catches extreme rain**: Nov 2024, 332 mm vs 500–800 mm published in the mountains.
- Tide data **only from 2023** — cannot be a training feature.
- Short test window: only risk level 1 has ≥ 30 **positive days** (31).

— CORRECTIONS TO REPORT 1

Measured, then corrected

REPORT 1 SAID

CORRECTED TO

~~42-year discharge record~~

29 years, 1997–2026

~~Archived forecast rain as input~~

Sensitivity test on perturbed rain

~~Target: official alert stage~~

Risk levels from discharge percentiles

~~Single-day rain beats accumulations~~

3-day accumulated rain is strongest ($r = 0.863$)

Official stages confirmed from Decision 05/2020: 1.0 / 2.0 / 3.5 m. Thank you – questions welcome.